Biomedical Signal Processing and Control 104 (2025) 107511

Contents lists available at ScienceDirect

Biomedical Signal Processing and Control

journal homepage: www.elsevier.com/locate/bspc

Cross-subject emotion recognition with contrastive learning based on EEG signal correlations

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ARTICLE INFO

Keywords:
Cross-subject emotion recognition
Signal correlation
Contrastive learning

ABSTRACT

In the field of cross-subject emotion recognition using electroencephalogram (EEG) signals, significant challenges arise due to substantial inter-individual differences and the complexities of EEG feature extraction. To address these issues, this paper introduces a novel contrastive learning method based on cross-subject signal correlation (CL-CS). The proposed method aligns EEG signals from different subjects across time, space, and frequency domains, effectively mitigating cross-subject data discrepancies and addressing the intricate dynamic characteristics of EEG signals in emotion classification tasks. The core of CL-CS is its contrastive loss function, which generates positive and negative sample pairs by calculating the correlations between EEG signals. This function maximizes the correlation of signal representations under identical emotional stimuli while minimizing it for signals associated with different stimuli. Such an approach facilitates the learning of robust and invariant emotional representations across subjects, enabling a more comprehensive capture of similarities in EEG data between individuals. In the emotion classification stage, the parameters of the three-domain encoder learned during the contrastive learning phase are used to extract differential entropy (DE) features. A multi-layer perceptron classifier is then employed to predict emotions. The proposed method is experimentally validated on publicly available datasets, including FACED, THU-EP, and SEED. It achieves state-of-the-art performance in binary, ternary, and nine-class classification tasks. Moreover, the model's generalization ability is demonstrated by its successful application to unseen subjects and novel emotional stimuli. The code of CL-CS is publicly available at https://github.com/VCMHE/CL-CS.

1. Introduction

Emotion is the integrated experience of an individual's psychological and physiological responses to internal and external stimuli. Affective computing utilizes subjective experiences, physiological signals (such as skin conductance, electroencephalogram (EEG), heart rate, and respiration rate), as well as external expressions (auditory and visual signals) to assess the user's emotional state, aiming to achieve more natural human–computer interaction [1]. Despite the valuable information provided by subjective reports, issues of credibility and verifiability exist. Physiological signals, particularly EEG, effectively reduce subjective interference due to direct control by the autonomic nervous system and endocrine system. Numerous studies have shown its superiority in emotion classification accuracy over other physiological signals, believed to more accurately reflect individual emotional responses [2]. Both domestic and foreign researchers have made significant advancements in EEG-based emotion recognition, achieving promising results in emotion recognition using both traditional machine learning methods [3–5] and deep learning methods [6–8]. Xu et al. [9] proposed an impulse spiking neural network (SNN)-based emotion recognition method that converts EEG data into a 2D frame format, demonstrating excellent performance and computational efficiency. Similarly, Lin et al. [10] introduced a fusion model that leverages the strengths of 1D convolution and EGAT to capture the brain's topology for emotion classification, effectively reducing computational costs. While recent advancements in EEG-based emotion recognition have achieved notable results in terms of model accuracy and computational efficiency, challenges persist in cross-subject emotion recognition. Compared to single-subject scenarios, the accuracy of cross-subject emotion recognition declines significantly under the same classification model due to factors such as individual differences, personalized experiences, and variations in emotional tendencies. For instance, Kim et al. [11] explored a bimodal data fusion approach and employed Linear Discriminant Analysis (LDA) for emotion classification. However, their method achieved a maximum recognition accuracy of only 55% across three subjects, far lower than the 92% accuracy obtained for single-subject data. Similarly, Zhu et al. [12]

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https://doi.org/10.1016/j.bspc.2025.107511
Received 20 May 2024; Received in revised form 19 December 2024; Accepted 2 January 2025
Available online 10 January 2025
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utilized differential entropy as the emotion-related feature and a linear SVM classifier. They evaluated the recognition performance under both within-subject and cross-subject experimental settings. While the within-subject approach achieved an average recognition accuracy of 90.97%, the cross-subject recognition accuracy was considerably lower at 64.82%.

Challenges persist in the domain of cross-subject emotion recognition based on EEG, arising from individual differences and intricate feature extraction [13,14]. The acquisition of EEG datasets is time-consuming, while the calibration of individual training data is labor-intensive. Limited availability of specific subject EEG data in publicly accessible datasets hampers the support for large-sample methodologies. Individual differences, influenced by physiological factors such as age, gender, and life experiences, pose challenges in neglecting them and contribute to the non-stationarity of EEG signals, thus complicating the capture of individual characteristics across different subjects [15]. Consequently, the development of emotion recognition approaches with cross-subject generalization holds paramount importance for real-world applications involving new users and stimuli.

The accuracy of cross-subject emotion recognition under the same classification model significantly decreases compared to single-subject emotion recognition, attributed to the influence of factors such as personalized experiences and individual tendency disparities. Consequently, the model performance remains suboptimal [16]. To mitigate the individual variances in EEG signals and enhance the extraction of invariant emotional representations, researchers have recently explored the application of transfer learning methods in cross-subject emotion recognition [17,18]. This endeavor, rooted in domain adaptation and domain generalization principles, entails the consideration of labeled and unlabeled individuals, categorized as source and target domains, respectively. On one hand, endeavors have been made to minimize the distribution variance between the source and target domains [19,20], approximating the assumption of independent and identical distributions, thereby improving recognition performance in the target domain. On the other hand, endeavors aim to identify domain-invariant representations from the source domain [21,22]. Typically, models are trained across multiple source domains to facilitate superior generalization to novel domains, rather than being constrained to a singular target domain. While transfer learning holds promise in addressing data scarcity and generalization issues in cross-subject emotion analysis tasks, the notable discrepancies in EEG signals across distinct stimuli and users pose significant challenges. Notably, not all knowledge can be seamlessly transferred, limiting the efficacy of transfer learning methods. Furthermore, simplistic machine learning approaches demonstrate minimal consideration for the neuroscientific underpinnings of human emotion processing.

Inspired by neuroscientific research on inter-subject correlation (ISC) [23], which suggests that brain neural processing patterns are shared among subjects when undertaking similar tasks or facing similar stimuli, numerous studies have integrated contrastive learning to acquire invariant emotion representations among subjects to address individual differences. Leveraging the advantage of contrastive learning, which typically requires less annotated data, effectively addresses the limitation of limited EEG data. However, directly applying contrastive learning frameworks from computer vision or natural language processing to the EEG emotion analysis domain has neglected the characteristics of emotion analysis based on EEG. There are significant differences in feature representation between EEG signals and image or text data. Additionally, EEG signals are time-series data, unlike static image or text data. Factors such as time, frequency, electrode placement, and number of acquisition channels all have different effects on the information contained in EEG signals [24]. Simply transferring existing contrastive learning frameworks is insufficient to capture and utilize the complex dynamic characteristics of EEG signals. Adaptive adjustments to the methods are needed to fully exploit the spatiotemporal information of EEG data.

The multidimensional features of EEG signals are crucial for downstream classification tasks. EEG signals are recorded from different locations on the scalp through multiple electrode channels, each reflecting the electrical activity of the brain region beneath it [25]. Different brain regions correspond to different functions, such as movement, sensation, vision, and emotion. Using multiple channels can obtain information on the electrical activity of different brain regions, thereby more accurately locating and analyzing brain activity patterns. EEG signals typically include rhythmic oscillations of various frequencies, whose frequency characteristics reflect the brain's activity levels in different states [26]. Different frequency brainwaves are associated with different types of neural activity. Additionally, the temporal characteristics of EEG signals capture the dynamic changes in brain activity, reflecting the brain's responses at different time points [27].

Building upon these multidimensional features, this paper proposes a contrastive learning method based on inter-subject signal correlation (CL-CS), which comprehensively considers the channel, time, and frequency dimensions of EEG signal features to better understand the relationship between brain activity and emotion classification. It also considers the amplitude and phase information of EEG signals to more comprehensively capture the correlation between signals. Specifically, the research framework of this paper sets up two stages based on the contrastive learning process [28], namely the invariant representation learning stage and the emotion prediction classification stage. During the contrastive learning process, the 3-domain encoder based on the convolutional neural network (CNN) learns invariant emotional feature representations of EEG signals across electrode channels, time, and frequency, and uses an improved contrastive loss function based on signal correlation to maximize the correlation of signal representations under the same emotional stimulus (positive pairs) while minimizing the correlation between signals under different stimuli (negative pairs), constructing a contrastive learning model more suitable for EEG signals. In the prediction process, the DE features of the signals are extracted using the 3-domain encoder parameters learned during the contrastive learning stage, and a multilayer perceptron classifier is used to recognize the emotions of the subjects.

This paper contributes in three main aspects:

(1) By considering the intricate dynamic characteristics of EEG signals and amalgamating feature insights from multiple dimensions including EEG signal channels, time, and frequency, we enhance our understanding of the relationship between brain activity and emotion classification. We achieve this through a 3-domain encoder for aligning multidimensional EEG time series data and an effective representation projector for learning invariant emotion representations among different subjects. This minimizes inter-subject data discrepancies, thereby enhancing the generalization and practicality of the emotion recognition model.

(2) Diverging from traditional contrastive learning frameworks, CL-CS formulates a contrastive loss function by computing the correlation between EEG signals. Compared to the conventional cosine similarity calculation method in traditional contrastive loss functions, the signal correlation coefficient considers both the amplitude and phase information of the signals, comprehensively capturing signal similarities and exhibiting greater tolerance to noise. This enables more accurate similarity measurement for multidimensional signals, making it better suited for downstream emotion classification tasks.

(3) We apply the 3-domain encoder parameters learned in contrastive learning to downstream differential entropy (DE) feature extraction and specific classification tasks. By combining contrastive learning and joint learning methods, we effectively enhance data utilization and achieve proficient cross-subject emotion classification without requiring large amounts of data. This improves the performance and generalization ability of the CL-CS model in classification tasks, while better utilizing task correlations.

2. Related work

EEG is a highly complex dynamic signal with non-stationarity, low signal-to-noise ratio, and uncertainty in time, space, and frequency dimensions, which makes it challenging to efficiently extract stable and useful features. EEG signals vary continuously with emotional states, individual differences, and environmental factors, making it difficult for traditional feature extraction methods to comprehensively capture the affective information in them. Cross-subject feature extraction is even more challenging. Significant differences in brain structure, emotional responses, and signal amplitude across individuals make it difficult to use the same feature space across subjects. This dynamic complexity is particularly salient in emotion classification tasks, where the model not only needs to extract stable features in the spatio-temporal frequency domain, but also needs to be robust to cross-subject individual differences. To cope with these problems, researchers have used various strategies to enhance the model's ability to adapt to dynamic properties, for example, Zhong et al. [29] proposed Regularized Graph Neural Networks (RGNN), which takes into account the biological topology of different brain regions, and employs two regularization methods to capture local and global inter-channel relationships in EEG signals for feature extraction. FAWTEEG [30] delves into the channel-specific features of EEG signals, decomposes them into different sub-bands using FAWT, and extracts emotional representations from these sub-bands for emotion recognition. EEANett [31] integrates the interactions between other modal features and EEG signal features into feature learning, and exploits the mechanism of multi-head self-attention to enhance multimodal feature extraction for emotion recognition. TC-Net [32] Uses transformer to extract EEG emotion representations and uses capsule networks to refine the emotion representations for emotion recognition. CapsNet [33] combines the temporal and spatial domain information of the EEG signals to incorporate the signals into a sparse spatio-temporal framework so that it can be easily processed with deep learning algorithms in the computer domain by training a CapsNet architecture for feature extraction for emotion recognition. However, since the voltage amplitude of the EEG signals varies greatly between subjects and the activity level of the EEG signals also varies from person to person, this individual difference problem makes the emotion recognition model perform poorly on unseen subject data, and the generalization remains poor.

In order to further enhance the generalization of the model and overcome the problem of cross-subject individual differences, researchers have attempted to apply methods such as Transfer Learning and Comparative Learning [34] to cross-subject emotion recognition. Migration learning starts from Domain Adaptation [35] and Domain Generalization [36], considering both labeled and unlabeled individuals (source and target domains) to mitigate the impact of individual differences across subjects on model performance. The objective of domain adaptation (DA) is to enable a model trained on the source domain to perform effectively in the target domain. This typically involves addressing the domain shift problem between the source and target domains, allowing the model to adapt to differing data distributions. Various domain adaptation techniques have been employed. MTL-MSRN [37] utilizes multi-scale residual neural networks to capture the interactions between different brain regions in EEG signals, applying meta-transfer learning to minimize individual differences among subjects for emotion recognition. EEDA [38] adopts an end-to-end approach to jointly optimize the feature extractor, domain discriminator, and classifier, utilizing information from the source domain to accurately obtain correct classification in the target domain. Chai et al. [39] proposed a strategy called Adaptive Subspace Feature Matching (ASFM), which integrates marginal and conditional distributions into a unified framework to reduce the differences in marginal and conditional distributions between the source domain and the unlabeled target domain. This effectively reduces domain discrepancies and mitigates performance degradation, achieving a certain level of generalization. However, such emotion recognition methods are impractical for real-world applications as they rely on access to large amounts of data from subjects. Furthermore, domain generalization (DG), which primarily focuses on how to apply knowledge learned from multiple source domains to an unknown target domain without requiring specific target domain labels, has been introduced into EEG emotion recognition. Li et al. [40] applied DG methods to the emotion recognition domain. By leveraging knowledge from multiple source domains to learn how to generalize to unseen target domains, new possibilities for training universal models are provided. Blanchard et al. [41] proposed a novel adversarial domain generalization framework(DResNet), which utilizes domain information to learn domain-shared weights and domain-specific weights. The domain-shared weights are used to classify different subjects. Zhao et al. [42] divided EEG representations into subject-specific private components and shared emotional components universal to all subjects, proposing an autoencoder architecture to learn domain-shared and domain-private emotional representations. DG methods help enhance the robustness of the model, making it more resistant to data changes and uncertainties, which has some effect on EEG signals with non-stationarity and high variability. However, apart from insufficient data, some DG methods may lead to a decrease in model interpretability, making it difficult to understand how the model analyzes and predicts emotions of different subjects.

Due to the domain shift problem caused by the mismatch between the data distributions of the source and target domains in transfer learning, some researchers have begun to apply contrastive learning frameworks from computer vision or natural language processing to the field of electroencephalogram (EEG) emotion analysis. This involves creating contrastive sample pairs from unlabeled data to train models and creating large-scale self-supervised labels or data representations for downstream emotion classification tasks by comparing the correlation or irrelevance between aligned samples from different subjects. Xie et al. [43] proposed a self-supervised model that automatically learns advanced representations of EEG by learning six different EEG transformations. Mohsenvand et al. [44]introduced a framework based on contrastive learning to learn representations from EEG signals, enhancing combinations of data from multiple channels, and extending the SimCLR framework to time series data to train channel feature extractors, effectively facilitating EEG-based emotion classification tasks. Banville et al. [45] explored two tasks based on temporal context prediction and encoded contrastive predictions for clinical relevant issues related to EEG-based sleep staging and pathology detection, confirming that the pre-training process can reveal the underlying structure of effects, gender effects, and pathological effects. Although the aforementioned works have to some extent addressed the issue of limited labeled EEG data, inter-subject variability has not been effectively mitigated.

As conclusions from Inter-Subject Correlation (ISC) studies emerge, contrastive learning has made new strides in the field of cross-subject affective analysis. ISC involves aligning brain activity data from multiple individuals for comparison, and then quantifying ISC by computing the correlation or similarity of brain activity data between different individuals to determine if consistency exists. Shen et al. [46] proposed the CLISA method, which constructs subject-invariant emotion representations by achieving alignment of spatiotemporal representations among subjects, effectively reducing inter-subject data differences. However, it still overlooks the complex dynamic characteristics of EEG signals; simply performing temporal operations on frequency-domain naive EEG signals may lead to loss of important information, and employing static data contrastive learning methods may be biased. Inspired by the aforementioned work, this paper systematically investigates the data alignment problem of EEG signals in the frequency, time, and spatial dimensions, considering the importance of feature representations in downstream classification tasks across these three dimensions. Frequency-domain features can provide crucial information about brain activity, aiding in identifying different states and cognitive

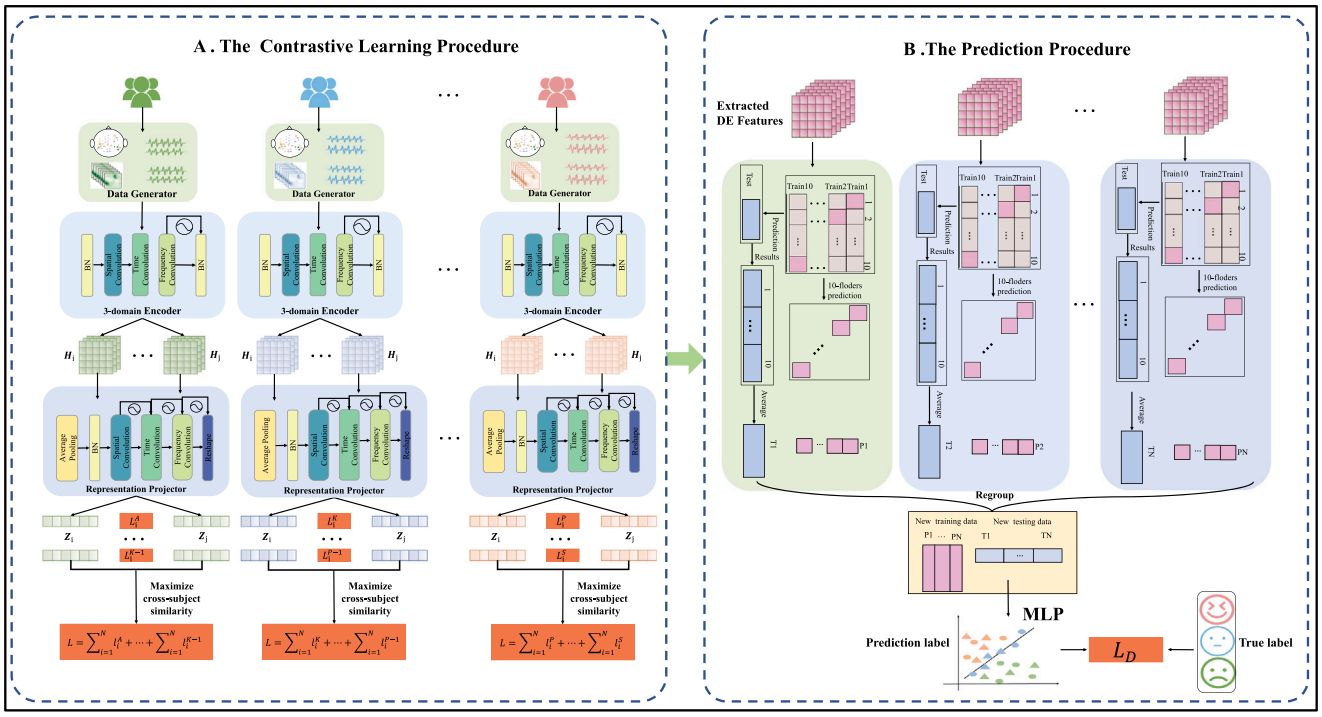


Fig. 1. The illustrates the overall model framework of the CL-CS method. The left section pertains to the contrastive learning phase, while the right section corresponds to the predictive classification phase, the models were classified by ten-fold cross-validation.

processes. Time features capture the dynamic changes in brain activity, analyzing the brain's responses at different time points. Spatial features help understand the interactions and connections between different brain regions, crucial for studying the synergy between brain areas. In this study, achieving data alignment among different subjects comprehensively considers alignment across these three feature dimensions to enhance the performance of affective classification tasks and provide a comprehensive understanding of brain activity. Furthermore, this paper constructs a contrastive loss function by computing the correlation between EEG signals. The signal correlation coefficient considers both the amplitude and phase information of signals, providing a more comprehensive capture of similarity between signals compared to traditional cosine similarity calculation methods in contrastive loss functions. It is also more tolerant to noise in signals, enabling more accurate similarity measurement for multidimensional signals, thereby improving the effectiveness of contrastive learning and its application in downstream affective classification tasks.

3. Methods

This section focuses on our proposed comparative learning method (CL-CS) based on subject signal correlation. The overall framework of the proposed method is shown in Fig. 1, which consists of two stages: invariant representation learning and prediction classification. In the invariant representation learning phase, we divide the positive and negative sample pairs by a data generator, and then use a 3-domain encoder and a representation projector to learn the inter-subject aligned data representations. The label prediction phase extracts the EEG invariant representations by using the 3-domain encoder trained in the invariant representation extraction phase and performs differential entropy(DE) features [47] extraction, and then performs emotion label prediction and classification by using the obtained DE features.

CL-CS is based on the research theory that 'when multiple subjects perform similar tasks or face similar stimuli, the neural processing patterns of the brain are shared among them', i.e., the EEG signals of different subjects under the same stimulus are similar, and with reference to the computational method of ISC, we aligned the data of different subjects' brain activities in the time, frequency, and channel domains, and the data are aligned in the time and frequency domains. and frequency domains, and channels for comparison, and then quantify the correlation or similarity of brain activity data between different individuals through comparative learning to determine whether there is consistency, and then learn similar invariant affective representations between individuals. The explanation of the symbols in the paper design is shown in the Table 1.

Table 1
Summary of major notations.

Notation	Description
SpatialConv	Spatial domain encoder convolution operation
TemporalConv	Time domain encoder convolution operation
FreqConv	Frequency domain encoder convolution operation
SpatialConv2	Spatial domain representation projector convolution operation
TemporalConv2	Time domain representation projector convolution operation
FreqConv2	Frequency domain representation projector convolution operation

3.1. Invariant representation learning phase

The invariant representation learning phase consists of four modules: the data generator, the 3-domain encoder, the representation projector, and the contrast loss calculation. Firstly, the data generator divides the input EEG data into several small batches and generates several pairs of positive and negative samples for training, then these small batches are put into the 3-domain encoder, which uses spatial convolution operator, temporal convolution operator, and frequency convolution operator to process these segments to realize subject data alignment, and finally the aligned invariant representations are mapped to another latent space through the projector to calculate the correlations of EEG signals across subjects. The model is trained with the goal of minimizing contrast loss, and the parameters of the 3-domain encoder and projector are optimized as the training process progresses to learn invariant affective representations across subjects, yielding optimal parameters for the prediction classification stage.

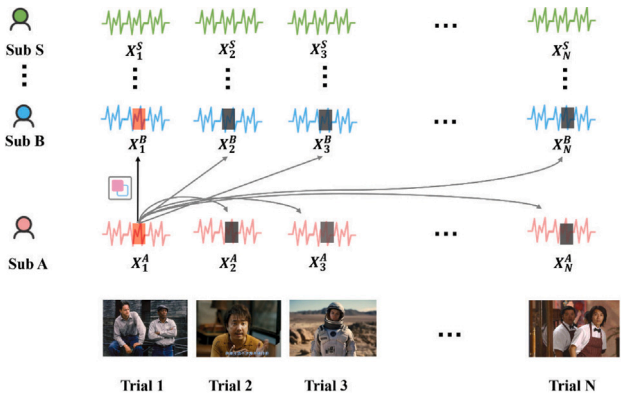


Fig. 2. The illustrates the process of partitioning positive and negative sample pairs during the contrastive learning stage. As depicted in the figure, X_i^A and X_i^B which derived from the same video stimulus, are paired as positive samples, while the remaining samples paired with X_i^A constitute the negative sample pairs.

3.1.1. Data generator

The main focus of the invariant representation learning phase in this approach is to differentiate identical emotional representations among EEG signals from different subjects under the same stimulus (environment). The data generator is primarily responsible for partitioning the input EEG signals into multiple mini-batches and delineating positive and negative sample pairs for the contrastive learning process during this phase.

The standard protocol for EEG data acquisition entails the collection of EEG data from numerous participants across N trials. During each trial, participants are exposed to emotionally stimulating video stimuli. Each participant undergoes N trials under various emotional stimuli conditions, with EEG data being recorded. Since EEG signals usually take a few seconds to stabilize after emotional induction to reflect the complete emotional state, we referred to the processing methods of THU-EP and FACED datasets and adopted a similar truncation strategy. In our experimental analyses, we selected the last 30S seconds of EEG data as the study object to ensure the continuity and validity of the emotional state. This strategy not only enhances the correspondence between emotion labels and EEG data, but also helps to reduce the computational complexity and avoid processing long and redundant signal segments. In total, each participant contributes N samples. For example, we take N samples from subject A, which we define as $X_1^A, X_2^A, \dots, X_N^A$ $(X_i^A \in \mathbb{R}^{M\times T})$, M represents the number of channels used for EEG signal acquisition, T denotes the number of time points in an EEG sample, indicating the quantity of available data time points in that sample. Similarly, the N samples taken from subject B are defined as $X_1^B, X_2^B, \dots, X_N^B$ $(X_i^B \in \mathbb{R}^{M\times T})$, Where samples$X_i^A$ and X_i^B captures EEG data collected during the same temporal interval for the same stimulus across participants A and B. The data generator constructs each pair of samples from both participants into a minibatch, defining set $D = \{X_i^s \mid i = 1, 2, \dots, N; s \in \{A, B\}\}$. The data generator establishes positive and negative sample pairs within each of the divided mini-batches, designating two samples from the same stimulus as positive sample pairs, and the remaining as negative sample pairs (Fig. 2). Such as giving a sample X_i^A the corresponding sample X_i^B under the same stimulus segment forms a positive sample with it, and the remaining 2(N-1) samples $X_j^s \mid j = 1, 2, \dots, N; j \neq I; s \in \{A, B\}$ all form negative samples with X_i^A.

3.1.2. 3-domain encoder

Utilizing paired EEG signals such as X_i^A and X_i^B as input, the 3-domain encoder employs spatial, temporal, and frequency convolutional operators to perform signal transformation, ensuring alignment between signals. This approach achieves three-dimensional alignment of data across time, frequency, and channels, while concurrently learning invariant emotional representations across subjects.

EEG data acquisition typically involves capturing activity from different brain regions through multiple channels. Data from a single channel may stem from various stimulus environments or even different subjects [48]. The 3-domain encoder of the CL-CS model initially utilizes spatial convolutional operators to transform signals into the corresponding latent space for signal tracing. It aligns input data along the spatial dimension (i.e., channels) to ensure uniformity and captures invariant emotional features and patterns. Spatial convolution is defined as:

$$H_{k_1}^{(1)} = SpatialConv(W_{k_1}, X), \quad k_1 = 1, 2, \dots, n \tag{1}$$

where $W_{k_1} \in \mathbb{R}^{1\times M}$ is the weight of the k_1 one-dimensional spatial convolution filter, a total of n spatial filters are set here, $H_{k_1}^{(1)}$ is the feature representation extracted by spatial convolution, Each row in $H_{k_1}^{(1)}$ is a potential feature representation determined by a linear combination of the original signal X_i^S, we simply expressed as $H^{(1)} = SpatialConv(X)$.

Considering the temporal variation of EEG signals, which reflect the dynamic characteristics of brain activity [49], it is essential to align them temporally to compare data from different subjects and under different stimulus conditions reasonably. This ensures the temporal consistency of events and allows for accurate comparison of events and signals across different data sources, without temporal shifts or misalignment. Additionally, CL-CS utilizes temporal convolutional operators via sliding windows to achieve signal alignment in the time domain and learn invariant representations of EEG signals in the temporal dimension. The temporal convolution is defined as:

$$\begin{aligned} H_{k_2}^{(2)} &= TemporalConv(V_{k_2}, H_{k1,t:t+P_1-1}^{(1)}), \\ &k_2 = 1, 2, \dots, m; \quad t = 1, 2, \dots, T \end{aligned} \tag{2}$$

where $V_{k_2} \in \mathbb{R}^{P_1}$ represents the weights of the k-th time domain convolution filters. The time domain convolution is here equivalent to a finite impulse response function with a filter length is P_1 and several filters is m. We populate the input $H_{k_1}^{(1)}$ to ensure that the output $H_{k_2}^{(2)}$ keeps the length Tconstant in the time dimension. We denote the time domain convolution operation as a whole and simply represented as $H^{(2)} = TemporalConv(H^{(1)})$.

EEG signals typically exhibit rhythmic oscillations across a spectrum of frequencies, where the frequency characteristics reflect the brain's activity levels across different states, and distinct types of neural activities are associated with varying frequency brainwaves [50]. CL-CS applies frequency convolution to EEG signals, accentuating frequency matching and similarity between two signals to achieve data alignment in the frequency domain, and further extracts invariant emotional representations of different EEG signals in the frequency domain. Frequency convolution is defined as:

$$H_{k_3}^{(3)} = M_{k_3} * H_{k_2,}^{(2)}, \quad k_3 = 1, 2, \dots, f \tag{3}$$

where $M_{k_3} \in \mathbb{R}^M$ represents the weight of the k-th frequency domain convolution, there are f frequency domain convolution kernels. We simply represent the frequency domain convolution operation as $H^{(3)} = FreqConv(H^{(2)})$.

3.1.3. Representation projector

The features learned by the 3-domain encoder from the same input minibatch are mapped to similar spatial positions, while the feature vectors from different inputs are mapped to different spatial positions. This mapping process brings similar samples closer together in the feature space, facilitating subsequent tasks to distinguish them more easily [51]. CL-CS employs convolutional units similar to those of the 3-domain encoder to compose the projector, eliminating the need to redesign the entire system. The model can adapt to different tasks and data without difficulty, and it can simultaneously process different data samples or features, thereby enhancing training efficiency.

The representation projector of CL-CS achieves spatial feature fusion of diverse latent signals through additional spatial convolutions to capture correlations among multiple inputs.

$$G = \sigma(SpatialConv2(\hat{H})) \tag{4}$$

The spatial convolution here has a similar computational process to the spatial convolution in the 3-domain encoder, but the difference is that here it is a depth spatial convolution with the filter size of $1 \times K_1$. The input $\hat{H}$ of the convolution layer has multiple feature graphs of size $K_1 \times \lfloor T/S \rfloor$ and each of the input feature graphs is processed by C spatial convolution filters, so there are a total of CK_2 filters. Utilizing extended time convolution to identify temporal patterns within signals.

$$N = \sigma(TemporalConv2(G)) \tag{5}$$

The temporal convolution here is similar to the temporal convolution in the 3-domain encoder, but the difference is that the depth temporal convolution here is the filter size of P_2. C temporal convolution filters process each input feature graph G to obtain C^2K_2 feature graphs.

By further applying frequency convolution, the main frequency components of the signal are identified.

$$Z = \sigma(FreqConv2(N)) \tag{6}$$

The frequency convolution here shares a similar calculation process as the frequency convolution in the 3-domain encoder, but the difference is that the depth frequency convolution here is the filter size of k_3. C frequency convolution filters process each input feature map N to obtain C^3K_3 feature maps. Then, the dimensionality of Z is transformed into a one-dimensional vector z, which facilitates the next step of correlation calculation.

3.1.4. Contrastive loss calculation

EEG signals have high-dimensional and complex time-series and frequency-domain properties, and cross-subject signal correlation helps to extract stable affective features from different time periods and frequency bands. Different from the limitation of matching individual signals under the traditional calculation of sample similarity, the introduction of signal correlation is more suitable for the nonlinear and heterogeneous data structure of EEG, which not only takes into account the amplitude of the signals, but also captures the phase information, which enables the model to learn the similarity between signals in a more comprehensive way. In addition, compared to the traditional cosine similarity metric, the correlation metric has stronger noise robustness, which is particularly suitable for signals like EEG data that contain multi-dimensional spatiotemporal information.

The input sample $D = \{X_i^s \mid i = 1, 2, \ldots, N; s \in \{A, B\}\}$ is converted to $D_{latent} = \{z_i^s \mid i = 1, 2, \ldots, N; s \in \{A, B\}\}$ after the 3-domain encoder and representation projector processing. Calculate the correlation between the input positive sample X_i^A and X_i^B

$$\rho(z_i^A, z_i^B) = \frac{(Cov(z_i^A, z_i^B))}{Std\,Dev(z_i^A) Std\,Dev(z_i^B)} \tag{7}$$

The purpose of the contrastive loss is to maximize the correlation between the two EEG signals in the positive pair. Referring to the SimCLR framework [28], we adopt the normalized temperature scale cross-entropy loss as the contrastive loss function.

$$l_i^A = -log\left[\frac{exp\left(\rho\left(z_i^A, z_i^B\right)/\tau\right)}{\sum_{j=1}^{N}\prod_{[j\neq i]} exp\left(\rho\left(z_i^A, z_j^A\right)/\tau\right) + \sum_{j=1}^{N} exp\left(\rho\left(z_i^A, z_j^B\right)/\tau\right)}\right] \tag{8}$$

By minimizing the contrastive loss function, the model will increase the correlation between all positive sample pairs. Finally, the total loss for each mini-batch is defined as

$$\mathcal{L} = \sum_{i=1}^{N} l_i^A + \sum_{i=1}^{N} l_i^B \tag{9}$$

3.2. Label prediction phase

In the label prediction phase, we employ the 3-domain encoder parameters optimized during the invariant representation learning phase to convert the input EEG data into aligned data representations. Subsequently, we conduct differential entropy (DE) feature extraction on these representations for emotion label classification (Fig. 1).

In the Label prediction process, the data is defined as **X**, and the corresponding labels are defined as **Y**. Labels are used to describe or represent the sentiment information contained in the data and serve as categorical variables for sentiment analysis or sentiment classification tasks. For example, if there are three sentiment categories, the variable $y \in \{0, 1, 2\}$ is defined. During the Label prediction Phase, it is necessary to predict the sentiment label of each sample X^{pred}, corresponding to the final predicted output $\mathbf{Y}^{pre}$. This is achieved through the trained 3-domain encoder that transforms the input data **X**into aligned data representations.

$$H^{pred} = FreqConv(TemporalConv(SpatialConv(\mathbf{X}))) \tag{10}$$

Since EEG data acquisition usually uses multi-channel devices, EEG data are high-dimensional single samples, which are easy to overfit in classification models, and the computational cost and time complexity of high-dimensional data are higher. Therefore, we extract the DE features of the EEG signals that are widely used in sentiment classification tasks and the extraction of DE features is defined as:

$$H^{DE} = \frac{1}{2}\log(2\pi e\sigma^2\left(H^{pred}\right)) \tag{11}$$

where σ^2 represents the variance of the original EEG signal and H^{DE}, the value of differential entropy can be used to describe the irregularity or complexity of the EEG signal as it changes over different time windows, and the DE features help to understand the time-domain dynamics of the EEG signal. DE features were extracted from the output of the trained the 3-domain encoder. Subsequently, successive samples of DE features from one trial of one subject across time were concatenated and smoothed using a linear dynamical system (LDS) model. The resulting smoothed features were reshaped into one-dimensional vectors, serving as inputs to the classifier h^{DE}. Finally, a three-layer MLP was employed as a classifier to generate emotion prediction labels. The classification process of the classifier is defined as:

$$\mathbf{Y}^{pre} = MLP(H^{DE}) \tag{12}$$

4. Experimental

In the experiment, we compare the model with other existing cross-subject emotion recognition methods and evaluate its performance under limited label sample learning. By visualizing the feature representation learned by the model, the invariant emotion representation and its pattern between different subjects were explored. In addition, the rationality of the model design was further verified by the ablation experiment.

4.1. Dataset

To validate the effectiveness of our proposed model, experiments were conducted on three publicly available EEG emotion datasets: the SEED dataset [25], the THU-EP dataset [52], and the FACED dataset [53]. The details of the three data sets are shown in Table 2. In the SEED dataset, 15 subjects watched 15 movie clips inducing three emotions (negative, neutral, and positive), and the EEG signals of different emotional states were recorded by a 62-channel, 1000 Hz sampling rate ESI neuroscanning system. The THU-EP dataset contains EEG data from 80 subjects, a total of 28 video clips were selected in the dataset to trigger different emotional experiences, the datasets encompassed four negative emotions (anger, disgust, fear, and sadness), four positive emotions (amusement, inspiration, joy, and tenderness), along with neutral emotions. EEG signals were acquired using a 32-channel

Table 2
Datasets selected for model evaluation.

Dataset	SEED [25]	THU-EP [52]	FACED [53]
Emotional categories	3	9	9
Stimulation type	Video	Video	Video
Sampling rate (Hz)	200	250	250
Number of stimulating video clips	15	28	28
Number of subjects	15	80	123
Number of acquisition channels	62	32	32

Table 3
Hyperparameterization of the proposed model.

	Hyperparameters	THU-EP [52]	FACED [53]	SEED [25]
Base encoder	Spatial filter size M	30	30	62
	Temporal filter size P1	60	60	48
	Frequency filter size P1	60	60	60
	Num. of spatial filters K1	16	16	16
	Num. of temporal filters K2	16	16	16
	Num. of frequency filters K2	32	32	32
	Average pooling core length S	30	30	24
Projector	Spatial filter size K1	16	16	16
	Temporal filter size P2	6	6	4
	Frequency filter size P1	6	6	4
	Num. of spatial filters CK2	32	32	32
	Num. of temporal filters C2K2	64	64	64

wireless EEG system (NeuSenW32) at a sampling rate of 250 Hz. The FACED dataset recorded the 32-channel EEG signals from 123 subjects, essentially the dataset is an expansion of the THU-EP dataset, where each subject sequentially watched movie clips labeled with 28 emotions, and emotionally stimulating videos that similarly contained nine emotional categories.

4.2. Parameter setting

Regarding parameter settings, for the THU-EP dataset and the FACED dataset, each trial's time length is set to 30 s. Each sample's time length is established at 5 s, with a time step of 2 s. This results in a total of $(30-5)/2+1=13$ samples per trial. For the SEED dataset, each sample's time length is set to 30 s, with a time step of 15 s, resulting in an average of 13 samples per trial. The hyperparameters of the modeling approach for each dataset are detailed in Table 3. During the training process of contrastive learning, hierarchical normalization was applied to the inputs of the 3-domain encoder, the outputs of the average pooling, and the outputs of the spatiotemporal convolutions in the projector. This normalization involved connecting the same channels from different samples of a single subject in small batches in a z-score normalized manner [54,55]. The model was trained using the Adam optimizer with the training schedule set to 100 epochs. The initial learning rate was set to 0.0007, and the weight decay rate was set to 0.015. For the MLP classifier in the prediction process, two hidden layers were configured, each with 30 units. Rectified Linear Units (ReLUs) were employed between the two layers. The cross-entropy loss and the Adam optimizer were used to optimize the parameters, with the learning rate empirically set to 0.0005. The weight decay was chosen from values between 0.005, 0.011, 0.025, 0.056, and 0.125 through cross-validation.

4.3. Model evaluation

Model evaluation encompassed various aspects. Firstly, on the cross-subject emotion recognition task, 10-fold cross-subject cross-validation was performed on the THU-EP dataset and the FACED dataset. Leave-one-out cross-validation was utilized on the SEED dataset to assess the model's effectiveness in three-classification, nine-classification, and binary-classification tasks. Secondly, during training, experiments were conducted using 2/3 of the samples from the training dataset. In the testing phase, the remaining 1/3 of samples from the testing dataset were employed. This approach ensures that the stimulus in the test set has never been encountered by the model during training, validating the model's generalization under different stimulus environments. Simultaneously, it verifies whether the model genuinely learns cross-subject invariant emotion representations, avoiding overfitting to specific stimuli.

(1) Performance of the model in different classification tasks: To assess the cross-subject emotion classification effectiveness of the model in various tasks, we designed and implemented binary, three-class, and nine-class classification tasks. In the binary task, emotions were categorized as negative (anger, disgust, fear, sadness) or positive (entertainment, joy, inspiration, tenderness). The three-class task included positive, negative, and neutral emotions. The nine-class task covered specific emotions: entertainment, joy, inspiration, tenderness, emotion, anger, disgust, fear, sadness, and neutral.

(2) Model performance on different dataset: The model's performance was further evaluated on different datasets by extending experiments from the FACED dataset to include the THU-EP dataset and the SEED dataset. These datasets were collected under varying stimulus video conditions, with different numbers of subjects, acquisition channels, and types of emotions.

(3) Contrast experiment setup: To further validate the effectiveness of the CL-CS method, it was compared with several of the most competitive existing sentiment recognition methods, namely Differential Entropy (DE) features with MLP classifiers (denoted as DE+MLP) [56], Subspace Alignment (SA) [57], Correlation Component Analysis (CorrCA) [58,59], SeqCLR [44], DAN [60], DANN [61]and CLISA [62], GCB-Net [63], DGCNN [64], TSception [65], TCN [66], and AMDET [67] methods. Among them, De+MLP is a simple baseline method for sentiment classification without inter-subject comparisons. SA and SeqCLR achieve cross-subject sentiment recognition through unsupervised domain adaptation, SA and CorrCA utilize a transfer learning approach to achieve cross-subject inter-subject data alignment, SeqCLR achieves contrast learning through data augmentation, and CLISA utilizes contrast learning to achieve data alignment to improve emotion recognition performance, which is currently the optimal method under the contrast learning framework. GCB-Net design graph convolutional wide network to explore the deep information of EEG data, DGCNN method to dynamically learn inter-subject features, TSception to achieve cross-subject emotion recognition through multi-scale convolutional neural network, TCN to apply cross-modal knowledge distillation, AMDET to achieve cross-subject emotion recognition through multidimensional global attention mechanism.

5. Result and discussion

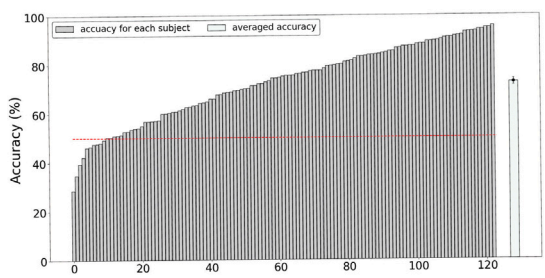
5.1. Performance on various dataset

5.1.1. Performance on the FACED dataset

As shown in Table 4, the CL-CS model performed a binary classification task of positive and negative emotions on the FACED dataset with an accuracy of 72.5±15.3%. The classification accuracy was significantly improved by 22% compared to the DANN model and 2.4% relative to the current state-of-the-art CLISA algorithm. This suggests that our proposed comparative learning approach is more effective in cross-subject emotion recognition performance. In the nine-category emotion classification task, the CL-CS model achieves an accuracy of 43.4±13.7%, an improvement of 8.5% compared to the baseline model(DE+SVM). Notably, our model outperforms other models in both the dichotomous and the nine classification tasks, indicating that the model has excellent analytical capabilities in different task or stimulus environments and strong generalization capabilities that can be successfully extended for use in completely new datasets. Each participant's classification accuracy and the average classification accuracy are shown in Fig. 3.

Table 4
Emotion recognition performance on the FACED dataset.

Methods	Binary		Nine-Class	
	Avg (%)	Std (%)	Avg (%)	Std (%)
DE+SVM [68]	69.5	16.0	34.9	10.7
DE+MLP [56]	70.2	11.2	35.1	10.3
DANN [61]	50.5	7.1	54.1	8.3
DAN [60]	53.4	8.3	37.2	9.1
GCB-Net [63]	56.5	5.2	–	–
DGCNN [64]	56.2	4.5	–	–
TSception [65]	61.9	8.8	–	–
TCN [66]	55.2	3.5	–	–
AMDET [67]	59.1	4.3	–	–
CLISA [62]	70.1	15.8	41.3	12.7
Ours	**72.5**	**15.3**	**43.4**	**13.7**



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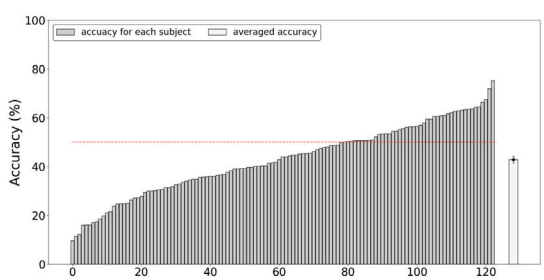

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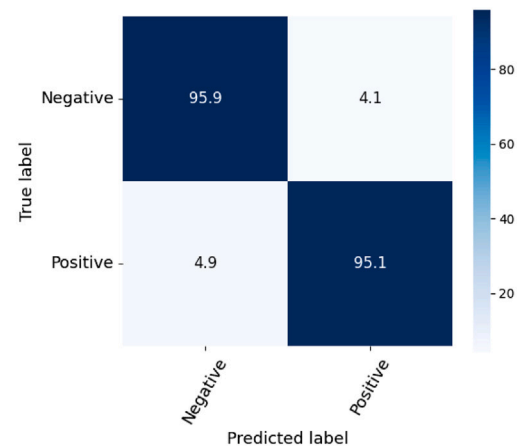
Fig. 3. The classification accuracies for each subject with CL-CS in the FACED dataset of (a) the binary classification task and (b) the nine-class classification task. The subjects are re-ranked according to their classification accuracies, increasing from left to right. The light gray bars indicate accuracies for each individual subject, and the light green bars indicate averaged accuracies across all subjects. The light green error bars represent standard deviations for all subjects. The red dashed line represents the probability level of the classification.



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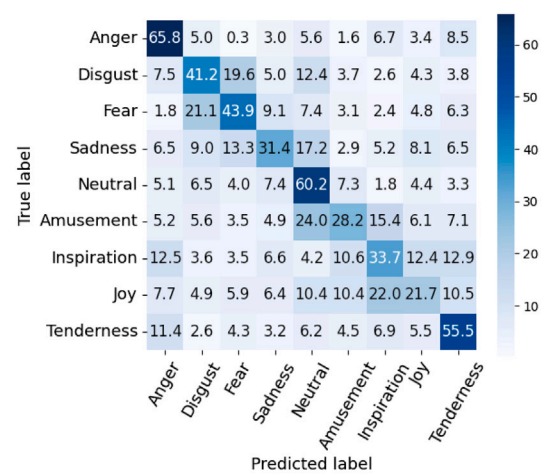

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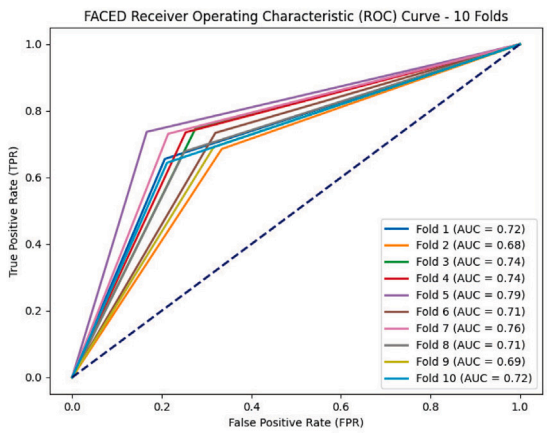
Fig. 4. Confusion matrices of different tasks on the FACED dataset. (a) The binary classification task confusion matrix of the FACED dataset. (b) Nine-classification task confusion matrix of the FACED dataset.

Based on the prediction results of the CL-CS model, we further conducted a detailed analysis of the confusion matrices between the emotion categories in the binary task and the nine-classification task (Fig. 4). We noted a high degree of accuracy in the recognition performance of the anger emotion in the nine-classification task, indicating a clear neural representation and significant cross-subject agreement for the anger emotion. In terms of positive emotions, we found that the recognition performance was best for the emotions of amusement and tenderness, for which the model was more capable of discriminating. We plot the ROC curves and PR curves of the ten-fold cross-validation of the model under the binary classification task, which further illustrates the effectiveness of the proposed method (Fig. 5).

5.1.2. Performance on the THU-EP dataset

The CL-CS model significantly improves accuracy when performing a binary sentiment classification task on the THU-EP dataset, achieving a binary sentiment classification accuracy of 95.4% ± 1.8%. Compared to the current state-of-the-art method, CLISA model, the accuracy was significantly improved by 23.5% and the standard deviation decreased by 1.1%. In addition, in the nine-category emotion classification task, the CL-CS model achieves a classification accuracy of 76.2% ± 14.9%, which exceeds the traditional cross-subject emotion classification method, and similarly improves by 30.5% compared to the current state-of-the-art CLISA model, with an increase in standard deviation, but within a manageable range. The results suggest that the features obtained through the CL-CS method are closer to a cross-subject invariant representation of emotion. Experiments show that CL-CS has good generalization and is suitable for complex cross-subject sentiment classification tasks. The excellent performance of the CL-CS method in EEG signal processing is mainly due to its ability to capture complex dynamic features in EEG signals. Specifically, the method constructs a contrast loss function based on signal similarity that takes into account the amplitude and phase information of the signals, captures the similarity between the signals in a more comprehensive way, is more tolerant to noise in the signals, and can more accurately perform similarity metrics for multidimensional signals, which can be better applied to downstream emotion classification tasks which makes it able to characterize the



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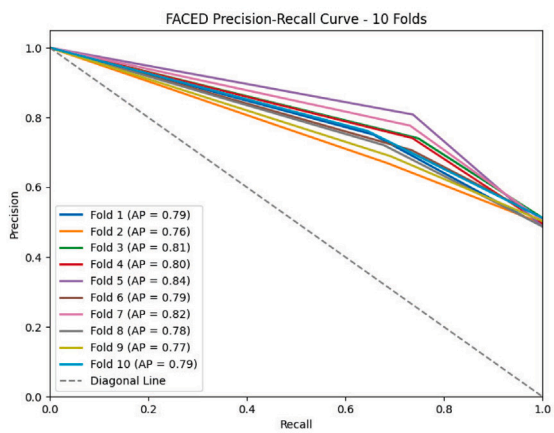
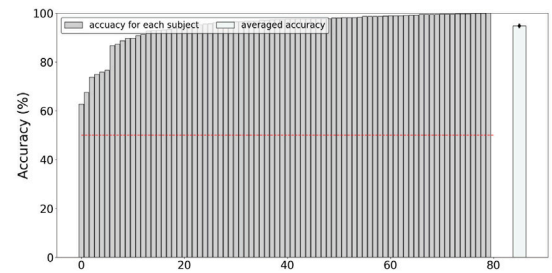


(b)

Fig. 5. ROC curves and PR curves of 10-fold cross-validation achieved by FACED dataset.

Table 5
Emotion recognition performance of different methods on the THU-EP dataset under different classification tasks.

Methods	Binary		Nine-Class	
	Avg (%)	Std (%)	Avg (%)	Std (%)
DE+MLP [56]	65.2	11.5	35.3	11.8
DAN [60]	54.5	9.3	37	10
DANN [61]	51.5	8.1	64.1	8.3
SA [57]	65.5	9.7	35.5	11.8
CorrCA [58]	64.5	11.0	34.5	10.4
SeqCLR [44]	64.0	9.8	34.3	10.5
GCB-Net [63]	55.4	4.3	–	–
DGCNN [64]	56.7	3.3	–	–
TSception [65]	59.1	5.9	–	–
TCN [66]	57.7	3.1	–	–
AMDET [67]	58.6	4.4	–	–
CLISA [62]	71.9	8.8	45.7	11.8
Ours	**95.4**	**7.7**	**76.2**	**14.9**



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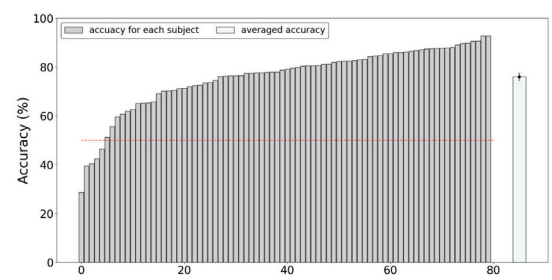

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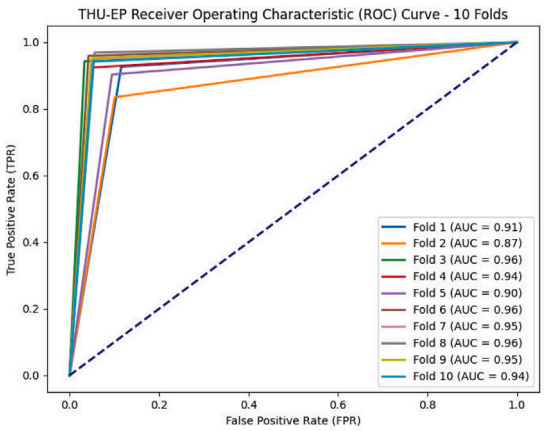
Fig. 6. The classification accuracies for each subject with CL-CS in the THU-EP dataset of (a) the binary classification task and (b) the nine-class classification task.

EEG signals in a more comprehensive way. In addition, compared to the CLISA method, which reduces frequency domain information to simple time domain alignment, the integration of the three dimensions of EEG signal channel, time, and frequency information provides a better understanding of the relationship between brain activity and emotion categorization, and is able to capture more detailed and more accurate cross-subject commonalities, which significantly reduces the inter-subject variability. The CL-CS method has also further validated the relationship between EEG signals with different frequencies and different types of emotional activities. The CL-CS method also further validated the strong correlation between different frequencies of EEG signals and different types of affective activities, which provides new ideas for further exploring the relationship between EEG signals and affective states. The comparison of the classification results between CL-SC and each method is shown in Table 5. In addition, we also cross-validated the ROC curves and PR curves of the models in the experimental binary classification task (Fig. 6) In addition, we further demonstrated the effectiveness of the proposed method by cross-verifying the ROC curve and PR curve of the model in the binary classification task during the experiment (Fig. 7).

Similarly, leveraging the model's predictive outcomes, we conducted an in-depth analysis of the confusion matrices for the THU-EP dataset in both binary and nine-class classification tasks, as depicted in Fig. 8. Upon scrutiny, we noted that in the context of the nine-class task, the model exhibited notably superior performance in recognizing positive sentiments compared to its proficiency in identifying negative sentiments. We employed t-Distributed Stochastic Neighbor Embedding(t-SNE) to visually map the sentiment difference feature space of the THU-EP dataset, as illustrated in Fig. 9. Notably, the sample distributions in the original dataset manifested a discernible and densely aggregated cluster structure, suggestive of shared attributes across specific dimensions. However, despite conspicuous differences, these attributes failed to elucidate the classification. In stark contrast, the DE feature sample space post-CLCS processing exhibited a rich tapestry of densely clustered samples distributed across various regions. This spatial arrangement indicated that these regions corresponded to high-density segments within the data distribution. It effectively underscored the similarity of samples proximal to these regions in certain features or attributes, revealing robust separability and interpretability. This distinctive feature space imbued the emotion recognition model with heightened interpretability and resilience. Experimental findings corroborate that the CL-CS method adeptly mitigates individual variances among subjects, thereby amplifying the efficacy of emotion



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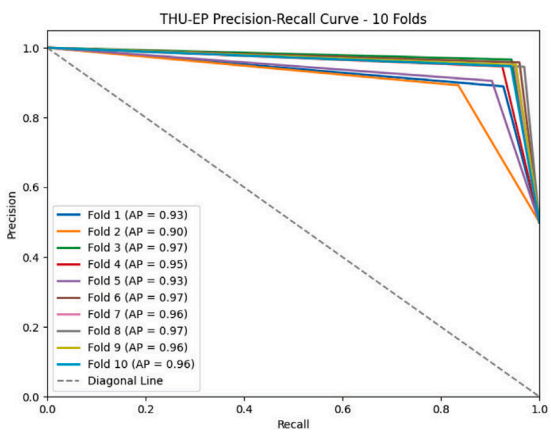

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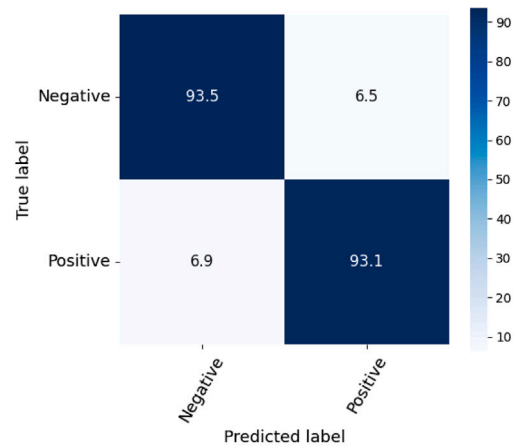
Fig. 7. ROC curves and PR curves of 10-fold cross-validation achieved by THU-EP dataset.



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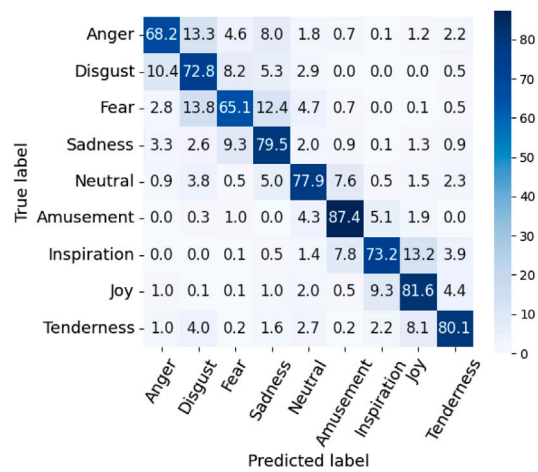


(b)

Fig. 8. Confusion matrices of different tasks on the THU-EP dataset. (a) The binary classification task confusion matrix of the THU-EP dataset. (b) Nine-classification task confusion matrix of the THU-EP dataset.

recognition performance.

5.1.3. Performance on the SEED dataset

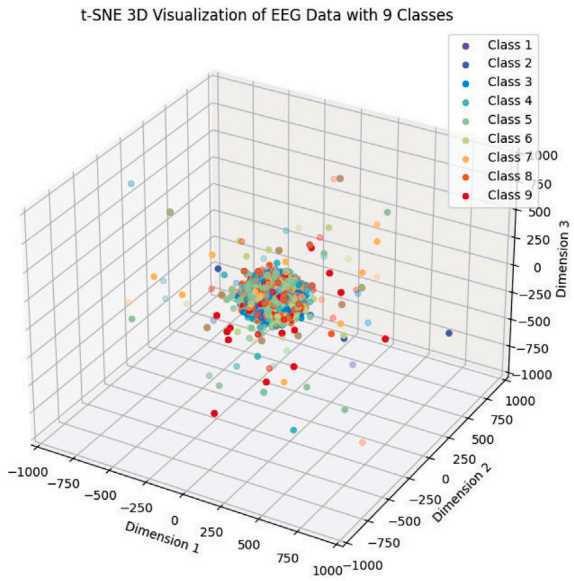
In our pursuit of assessing the efficacy of our method, we turn to the widely recognized SEED dataset. As shown in Table 6 the CL-CS model attains a noteworthy accuracy of 88.3% ± 8.9% in executing a three-categorical sentiment classification task that encompasses positive, negative, and neutral sentiments. Similar to our findings on the THU-EP dataset, the CL-CS model demonstrates a significant edge over alternative methodologies, including DE+MLP, SA, CorrCA, and SeqCLR, particularly in terms of predictive capabilities. Furthermore, the CL-CS model outshines previously acknowledged state-of-the-art models such as DAN, CorrCA, and CLISA, as reported in earlier studies. This underscores the substantive superiority of our comparative learning strategy when juxtaposed with domain adversarial approaches, thereby reaffirming the robustness and effectiveness of our proposed methodology. The confusion matrix of our model is depicted in Fig. 10. It can be observed that the model exhibits a higher tendency to misclassify negative emotions, particularly neutral emotions. Conversely, the classification accuracy for positive emotions is comparatively superior.

Table 6
Emotion recognition performance of different methods on the SEED dataset under the third classification tasks.

Methods	Three-class	
	Avg(%)	Std(%)
DE+MLP [56]	79.9	8.7
DAN [60]	83.8	8.5
DANN [61]	63.8	10.3
SA [57]	78.0	6.3
CorrCA [58]	73.7	10.2
SeqCLR [44]	78.4	9.2
GCB-Net [63]	68.4	12.0
DGCNN [64]	72.4	14.5
TSception [65]	66.2	18.1
TCN [66]	76.5	14.0
AMDET [67]	71.2	16.8
CLISA [62]	86.4	6.4
Ours	**88.3**	**8.9**

5.2. Statistical evaluation

In order to fully evaluate the performance of CL-CS, we perform a more in-depth analysis. Inspired by [69,70] we further compare the performance of multiple algorithms on the SEED dataset. Firstly, based on Table 7, we calculate the mean rankings of different methods, and the CL-CS method performs outstandingly among all models, with a



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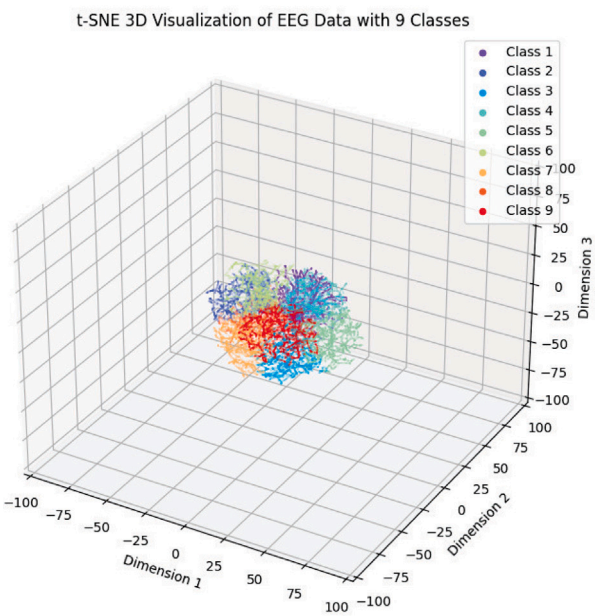
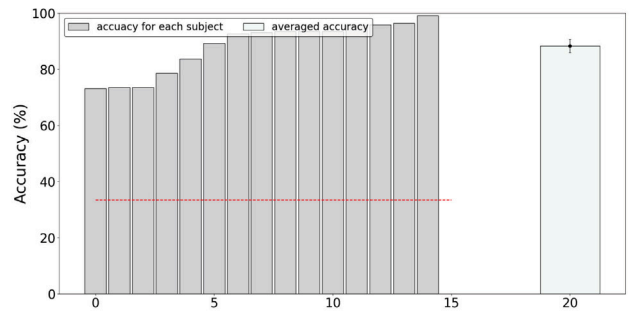


(b)

Fig. 9. (a) t-sne results of DE features for THU-EP raw data. (b) t-sne results of DE features extracted from the output of the 3-domain encoder after CL-CL model training.

mean ranking of 11.13, which is at the top of all methods. Compared to other methods, especially DANN (mean rank 2.00) and GCB-NET (mean rank 3.87), CL-CS has a significant advantage in cross-subject EEG signaling emotion recognition. Among these methods, DE+SVM and CLISA also performed better with 9.23 and 10.57, respectively, while SeqCLR and TCN followed with 8.33 and 7.67 rankings. These results indicate that although some traditional models are still competitive, CL-CS successfully enhances the accuracy and stability of sentiment recognition through improved contrast learning methods and cross-signal feature considerations. In addition, we calculate the Wilcoxon rankings of the different methods and show that the CL-CS method shows significant differences compared to other algorithms based on



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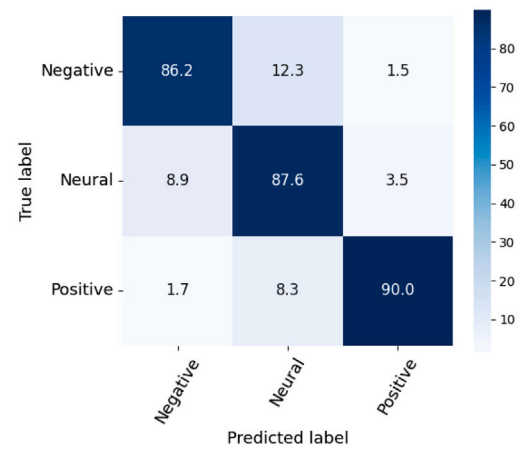

(b)

Fig. 10. (a) The third classification accuracy of the SEED dataset.(b)Third-classification task confusion matrix of the SEED dataset.

the p-values and hypothesis testing results of multiple methods. Specifically, the p-values of DE+SVM, DANN, SA, CorrCA, SeqCLR, GCB-NET, DGCNN, TSception, and TCN are all lower than 0.05, which indicates that their performance is significantly different from that of the CL-CS method, and therefore the hypothesis can be rejected that these algorithms are significantly inferior to CL-CS. The p-values of DAN and CLISA are 0.094604 respectively, which shows that there is no significant difference in performance with the CL-CS method. In summary, the CL-CS method demonstrates excellent performance in emotion classification, and successfully improves the accuracy and stability of the emotion recognition model through the improved contrast learning loss function and multiple considerations for cross-signal features.

5.3. Complexity analysis

In the task of cross-subject sentiment classification using contrast learning, the design of the loss function is critical to the performance and computational efficiency of the model. In this chapter, a complexity analysis of the proposed signal correlation-based construction versus the traditional sample similarity-based construction of the contrast loss function is presented. The computational complexity analysis for the traditional loss function can be summarized in several parts. Firstly, constructing a loss function based on sample similarity requires calculating the similarity between each pair of samples $sim(z_i^A),(z_i^B)$, assuming that the dimension of each sample is p, then computing the similarity between each pair of samples is usually O(P), and the complexity of computing the similarity of all pairs of samples is $O(N^2)$ for

Table 7
Mean Rankings and Wilcoxon Signed-Rank Test Results for Imbalanced Classification Methods vs. CL-CS (AUC and STD Metrics).

Method	Mean Rank	Wilcoxon p-value	Hypotheses
DE+SVM	9.23	0.006714	True
DAN	10.13	0.094604	False
DANN	2.00	0.000061	True
SA	8.20	0.00061	True
CorrCA	6.20	0.00061	True
SeqCLR	8.33	0.003357	True
GCB-Net	3.87	0.000427	True
DGCNN	5.67	0.000427	True
TSception	2.93	0.000061	True
TCN	7.67	0.003357	True
AMDET	5.07	0.000427	True
CLISA	10.57	0.094604	False
Ours	11.13	/	/

N^2 samples, where N is all possible pairs of samples. One exponential operation is performed for each sample pair, and for N sample pairs, $O(N^2)$exponential operations are performed, moreover, in the computation of the denominator, the complexity of both the product and the summation part is O(N), and the total complexity can be regarded as $O(N^2 * P)$. The CL-CS loss function relies on the calculation of correlation between signals, and the introduction of signal correlation provides a new perspective for the construction of the loss function. The complexity of signal correlation $\rho(z_i^A, z_i^B)$ in the similarity calculation is still $O(P)$, and thus for N samples, the total complexity of the similarity calculation is $O(N^2 * P)$Second, the exponential operation in the loss function also needs to be calculated $O(N^2)$ times calculations. In the calculation of the denominator, the complexity of both the product and summation parts is O(N), and the total computational complexity is also $(N^2 * P)$. However, since the second loss function adopts a similarity measure based on signal correlation, it can capture the intrinsic relationship between samples more effectively, thus reducing the impact of signal noise on the model performance to a certain extent, and providing stronger robustness and better performance in the EEG sentiment classification task.

5.4. Ablation experiment

To delve deeper into the plausibility of the CL-CS model in more detail, we conducted a series of ablation experiments that focused on three key steps of the CL-CS architecture: the data alignment module, the contrastive learning loss and the classifier. We choose the SEED dataset as the experimental benchmark and employ the same evaluation strategy as in the previous section. First, to validate the effectiveness of the data alignment module in the CL-CS model for cross-subject classification, we progressively remove the alignment steps in time, space, and frequency, and observe the changes in classification accuracy. This helps to quantify the contribution of different alignment steps in minimizing cross-subject differences. Second, we compare the variability between traditional contrastive learning loss computation methods and our proposed method by trying different contrastive learning loss computation methods, which helps to validate the effectiveness of the loss computation methods we adopt in optimizing the model and their effectiveness in combating cross-subject variability. Finally, the performance of the CL-CS model is further explored by replacing different classifier structures.

The results obtained from the ablation experiments on the SEED dataset shown in Fig. 11 effectively validate the importance of the three alignment steps, temporal, spatial, and frequency, in the data alignment module, suggesting that one of the three steps is indispensable. With the removal of each alignment step, the cross-subject categorization accuracies all showed a decrease. Specifically, removing the frequency dimension alignment step resulted in a 2.8% decrease in classification accuracy. This indicates that alignment in the frequency dimension

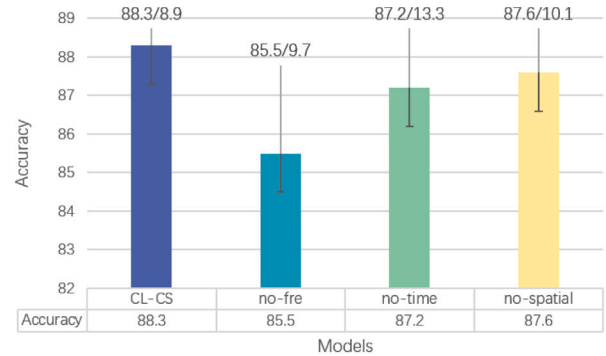

Fig. 11. Histogram comparing the accuracy/error of ablation experiments.

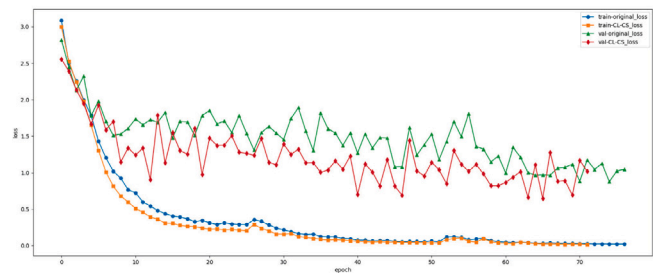

Fig. 12. Loss function convergence comparison plot.

significantly improves classification performance, suggesting that the frequency information of the signal plays an important role in cross-subject differences. Removing the temporal dimension alignment step resulted in a 1.1% decrease in accuracy, demonstrating that the temporal dimension is equally critical in removing subject differences. Finally, removing the spatial dimension alignment step resulted in a 0.7% decrease in final classification accuracy. These results indicate that the data alignment module largely reduces cross-subject differences, and we argue that the effectiveness of the data alignment module on the temporal, spatial, and frequency dimensions, especially the frequency dimension and the temporal dimension, underscores the importance of comprehensively aligning multidimensional data in emotion recognition or other related tasks. We experimentally compare the difference between the traditional contrastive loss calculation and the contrastive loss calculation based on the correlation between signals proposed in this paper, as shown in Fig. 12 The loss values obtained by the contrastive loss calculation method proposed in this paper are lower than the traditional contrastive loss in both the training and validation phases, and the model converges faster, which effectively verifies the reasonableness of the CL-CS model.

We replaced the classifiers in the CL-CS model and experimented with the effect of MLP [71], SVM [72], Logistic [73] and KNN [74] classifiers, respectively, as shown in the results of Fig. 13, the MLP classifier selected for the CL-CS model has significantly higher mean classification accuracy than the other classifiers on the SEED dataset.

5.5. Impact of key parameters on model performance

In this section, we evaluate the impact of learning rate and node dimension. The learning rate is a key parameter that determines the step size to minimize the loss function at each iteration. We obtained the best value from the range of 0.1 to 0.00001 by grid search. Table 8 shows the test results of the learning rate on the SEED dataset. It can be seen that CL-CS performs best on the SEED dataset with a learning rate of 0.0005. As the parameter increases from 0.1 to 0.0005, the accuracy of the model gradually improves, but as the parameter

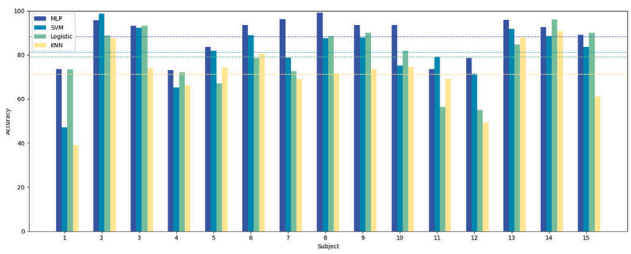

Fig. 13. Plot of classification accuracy for each subject under different classifiers.

Table 8
Performance Comparison of Different Learning Rates on the SEED Dataset.

Learning rate	Avg(%)	Std(%)
0.1	68.9	23.4
0.5	40.3	10.6
0.01	81.5	17.1
0.05	77.7	15.0
0.001	84.1	16.0
0.005	82.7	16.1
0.0001	84.8	16.0
0.0005	**88.3**	**8.9**
0.00001	81.0	12.8
0.00005	83.4	15.9

Table 9
Performance Comparison of Varying the Embedding Dimension on the SEED Dataset.

Dimension	Avg(%)	Std(%)
128	88.0	9.6
64	84.5	15.3
32	**88.3**	**8.9**
16	83.9	15.5
8	82.8	16.0
4	83.7	15.7

continues to exceed 0.0005, the model performance begins to decline. Too large a learning rate causes the model to converge to a sub-optimal solution. Node dimension is a key factor affecting model performance. We conducted an experimental validation on the SEED dataset to assess the impact of node dimensionality, as shown in Table 9 . As the feature dimension increases from 8 to 32, the model performance gradually improves. When the model performs best, the feature dimension is 32. when the dimension continues to increase, the noise introduced by the features leads to performance degradation.

6. Conclusion

The experiments above demonstrate that CL-CS achieves excellent classification performance across different datasets and classification tasks. By comprehensively integrating the multi-dimensional features of EEG signals, CL-CS enhances the understanding of the relationship between brain activity and emotion classification. The model shows strong generalization and versatility, contributing to the practical application of emotion recognition systems.

Despite the significant results obtained in cross-subject EEG emotion recognition, certain limitations remain. Specifically, the pre-processing of EEG signals from different datasets is complicated, and inconsistencies in the number of channels and signal duration can affect the model's generalization ability. These discrepancies may lead to information loss during feature extraction and signal processing, ultimately impacting the classification performance.

In future research, we will explore adaptive strategies based on channel correlation and signal dynamics to allow the model to dynamically adjust the input signal's duration and the number of channels during training. This approach aims to further improve the model's generalization, stability, and accuracy. Additionally, we plan to enhance the sharing of emotional categories and the distinguishable spatio-temporal patterns of EEG signals, which will help improve the interpretability of the emotion classification network architecture.

CRediT authorship contribution statement

Mengting Hu: Writing – original draft, Visualization, Validation, Software, Resources, Project administration, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Dan Xu:** Writing – review & editing, Supervision, Funding acquisition. **Kangjian He:** Writing – review & editing, Validation, Supervision, Project administration, Funding acquisition. **Kunyuan Zhao:** Writing – review & editing, Conceptualization. **Hao Zhang:** Writing – review & editing, Supervision.

Funding

This work was supported in part by the National Natural Science Foundation of China under Grant 62162068, 62202416, Grant 61761049, 62462066, in part by the Yunnan Province Ten Thousand Talents Program and Yunling Scholars Special Project under Grant YNWR-YLXZ-2018-022, in part by the Yunnan Provincial Science and Technology Department-Yunnan University *Double First Class* Construction Joint Fund Project under Grant No. 202301BF070001-025.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The authors do not have permission to share data.

References

[1] R.W. Picard, Affective Computing, MIT Press, 2000.
[2] X. Li, Y. Zhang, P. Tiwari, D. Song, B. Hu, M. Yang, P. Marttinen, EEG based emotion recognition: A tutorial and review, ACM Comput. Surv. 55 (4) (2022) 1–57.
[3] A. Mutawa, A. Hassouneh, Multimodal Real-Time patient emotion recognition system using facial expressions and brain EEG signals based on Machine learning and Log-Sync methods, Biomed. Signal Process. Control 91 (2024) 105942.
[4] S. Liu, Y. Zhao, Y. An, J. Zhao, S.-H. Wang, J. Yan, GLFANet: A global to local feature aggregation network for EEG emotion recognition, Biomed. Signal Process. Control 85 (2023) 104799, http://dx.doi.org/10.1016/j.bspc.2023.104799.
[5] M. Jafari, et al., Emotion recognition in EEG signals using deep learning methods: A review, Comput. Biol. Med. 107450 (2023).
[6] R. Fu, M. Cai, S. Wang, Y. Wang, C. Jia, Exploring non-Euclidean relationships in EEG emotion recognition: A dynamic graph attention network, Biomed. Signal Process. Control 94 (2024) 106276, http://dx.doi.org/10.1016/j.bspc.2024.106276.
[7] C. Fan, H. Xie, J. Tao, Y. Li, G. Pei, T. Li, Z. Lv, ICaps-ResLSTM: Improved capsule network and residual LSTM for EEG emotion recognition, Biomed. Signal Process. Control 87 (Part B) (2024) 105422, http://dx.doi.org/10.1016/j.bspc.2023.105422.
[8] M. Meng, J. Hu, Y. Gao, W. Kong, Z. Luo, A deep subdomain associate adaptation network for cross-session and cross-subject EEG emotion recognition, Biomed. Signal Process. Control 78 (2022) 103873, http://dx.doi.org/10.1016/j.bspc.2022.103873.
[9] F.F. Xu, D. Pan, H. Zheng, et al., EESCN: A novel spiking neural network method for EEG-based emotion recognition, Comput. Methods Programs Biomed. 243 (2024) 107927.
[10] X. Lin, J. Chen, W. Ma, et al., EEG emotion recognition using improved graph neural network with channel selection, Comput. Methods Programs Biomed. 231 (2023) 107380.
[11] J. Kim, Bimodal emotion recognition using speech and physiological changes, in: M. Grimm, K. Kroschel (Eds.), Robust Speech Recognition and Understanding, InTech Press, Rijeka, 2007, pp. 265–280.

[12] J.Y. Zhu, W.L. Zheng, B.L. Lu, Cross-subject and cross-gender emotion classification from EEG, in: Proceedings of the World Congress on Medical Physics and Biomedical Engineering, Springer Science and Business Media, Berlin, 2015, pp. 1188–1191.

[13] M. Kalashami, P. Pourhosein, S. Mir Mohsen, S. Hossein, EEG feature extraction and data augmentation in emotion recognition, Comput. Intell. Neurosci. 2022 (2022) 7028517, http://dx.doi.org/10.1155/2022/7028517.

[14] A.K. Singh, S. Krishnan, Trends in EEG signal feature extraction applications, Front. Artif. Intell. 5 (2023) http://dx.doi.org/10.3389/frai.2022.1072801.

[15] C. Wang, Y. Li, L. Wang, S. Liu, S. Yang, A study of EEG non-stationarity on inducing false memory in different emotional states, Neurosci. Lett. 809 (2023) 137306, http://dx.doi.org/10.1016/j.neulet.2023.137306.

[16] C. Ren, J. Chen, R. Li, W. Zheng, Y. Chen, Y. Yang, X. Zhang, B. Hu, Semi-supervised pairwise transfer learning based on multi-source domain adaptation: A case study on EEG-based emotion recognition, Knowl.-Based Syst. 305 (2024) 112669, http://dx.doi.org/10.1016/j.knosys.2024.112669.

[17] J. Quan, Y. Li, L. Wang, R. He, S. Yang, L. Guo, EEG-based cross-subject emotion recognition using multi-source domain transfer learning, Biomed. Signal Process. Control 84 (2023) 104741, http://dx.doi.org/10.1016/j.bspc.2023.104741.

[18] Y. Peng, H. Liu, W. Kong, F. Nie, B.-L. Lu, A. Cichocki, Joint EEG feature transfer and semisupervised cross-subject emotion recognition, IEEE Trans. Ind. Inform. 19 (7) (2023) 8104–8115, http://dx.doi.org/10.1109/TII.2022.3217120.

[19] W. Li, W. Huan, S. Shao, B. Hou, A. Song, MS-FRAN: A novel multi-source domain adaptation method for EEG-based emotion recognition, IEEE J. Biomed. Health Inf. 27 (11) (2023) 5302–5313, http://dx.doi.org/10.1109/JBHI.2023.3311338.

[20] Z. Li, E. Zhu, M. Jin, C. Fan, H. He, T. Cai, J. Li, Dynamic domain adaptation for class-aware cross-subject and cross-session EEG emotion recognition, IEEE J. Biomed. Health Inf. 26 (12) (2022) 5964–5973.

[21] Y. Gu, X. Zhong, C. Qu, C. Liu, B. Chen, A domain generative graph network for EEG-based emotion recognition, IEEE J. Biomed. Health Inf. 27 (5) (2023) 2377–2386, http://dx.doi.org/10.1109/JBHI.2023.3242090.

[22] J. Quan, Y. Li, L. Wang, R. He, S. Yang, L. Guo, EEG-based cross-subject emotion recognition using multi-source domain transfer learning, Biomed. Signal Process. Control 84 (2023) 104741.

[23] M.P. Hejnar, K.A. Kiehl, V.D. Calhoun, Interparticipant correlations: A model free FMRI analysis technique, Hum. Brain Mapp. 28 (9) (2007) 860–867.

[24] Y.-T. Lan, K. Ren, Y. Wang, W.-L. Zheng, D. Li, B.-L. Lu, L. Qiu, Seeing through the brain: image reconstruction of visual perception from human brain signals, 2023, arXiv preprint arXiv:2308.02510.

[25] W.-L. Zheng, B.-L. Lu, Investigating critical frequency bands and channels for EEG-based emotion recognition with deep neural networks, IEEE Trans. Auton. Ment. Dev. 7 (3) (2015) 162–175.

[26] Y. Ding, X. Hu, Z. Xia, Y.-J. Liu, D. Zhang, Inter-brain EEG feature extraction and analysis for continuous implicit emotion tagging during video watching, IEEE Trans. Affect. Comput. 12 (1) (2018) 92–102.

[27] N. Jrad, M. Congedo, Identification of spatial and temporal features of EEG, Neurocomputing 90 (2012) 66–71.

[28] T. Chen, S. Kornblith, M. Norouzi, G. Hinton, A simple framework for contrastive learning of visual representations, in: International Conference on Machine Learning, PMLR, 2020, pp. 1597–1607.

[29] P. Zhong, D. Wang, C. Miao, EEG-based emotion recognition using regularized graph neural networks, IEEE Trans. Affect. Comput. 13 (3) (2020) 1290–1301.

[30] V. Gupta, M.D. Chopda, R.B. Pachori, Cross-subject emotion recognition using flexible analytic wavelet transform from EEG signals, IEEE Sens. J. 19 (6) (2018) 2266–2274.

[31] Z. Yang, D. Li, F. Hou, Y. Song, Q. Gao, Deep feature extraction and attention fusion for multimodal emotion recognition, IEEE Trans. Circuits Syst. II (2023).

[32] Y. Wei, Y. Liu, C. Li, J. Cheng, R. Song, X. Chen, TC-Net: A Transformer Capsule Network for EEG-based emotion recognition, Comput. Biol. Med. 152 (2023) 106463.

[33] G.C. Jana, A. Sabath, A. Agrawal, Capsule neural networks on spatio-temporal EEG frames for cross-subject emotion recognition, Biomed. Signal Process. Control 72 (2022) 103361.

[34] Y. Zhang, Y. Liao, W. Chen, X. Zhang, L. Huang, Emotion recognition of EEG signals based on contrastive learning graph convolutional model, J. Neural Eng. 21 (4) (2024) 046060, http://dx.doi.org/10.1088/1741-2552/ad7060.

[35] C. She, C. Zhang, F. Fang, Y. Ma, Y. Zhang, Multisource associate domain adaptation for cross-subject and cross-session EEG emotion recognition, IEEE Trans. Instrum. Meas. 72 (2023) 1–12, http://dx.doi.org/10.1109/TIM.2023.3277985.

[36] D. Huang, S. Zhou, D. Jiang, Generator-based domain adaptation method with knowledge free for cross-subject EEG emotion recognition, Cogn. Comput. 14 (2022) 1316–1327, http://dx.doi.org/10.1007/s12559-022-10016-4.

[37] J. Li, H. Hua, Z. Xu, L. Shu, X. Xu, F. Kuang, S. Wu, Cross-subject EEG emotion recognition combined with connectivity features and meta-transfer learning, Comput. Biol. Med. 145 (2022) 105519.

[38] H. Zhao, Q. Zheng, K. Ma, H. Li, Y. Zheng, Deep representation-based domain adaptation for nonstationary EEG classification, IEEE Trans. Neural Netw. Learn. Syst. 32 (2) (2020) 535–545.

[39] X. Chai, Q. Wang, Y. Zhao, Y. Li, D. Liu, X. Liu, O. Bai, A fast, efficient domain adaptation technique for cross-domain electroencephalography (EEG)-based emotion recognition, Sensors 17 (5) (2017) 1014.

[40] Y. Li, H. Chen, J. Zhao, H. Zhang, J. Li, Benchmarking domain generalization on EEG-based emotion recognition, 2022, arXiv preprint arXiv:2204.09016.

[41] G. Blanchard, G. Lee, C. Scott, Generalizing from several related classification tasks to a new unlabeled sample, Adv. Neural Inf. Process. Syst. 24 (2011).

[42] M. Jiménez-Guarneros, G. Fuentes-Pineda, Learning a robust unified domain adaptation framework for cross-subject EEG-based emotion recognition, Biomed. Signal Process. Control 86 (2023) 105138.

[43] Z. Xie, M. Zhou, H. Sun, A novel solution for EEG-based emotion recognition, in: 2021 IEEE 21st International Conference on Communication Technology, ICCT, IEEE, 2021, pp. 1134–1138.

[44] M.N. Mohsenvand, M.R. Izadi, P. Maes, Contrastive representation learning for electroencephalogram classification, in: Machine Learning for Health, PMLR, 2020, pp. 238–253.

[45] H. Banville, O. Chehab, A. Hyvärinen, D.-A. Engemann, A. Gramfort, Uncovering the structure of clinical EEG signals with self-supervised learning, J. Neural Eng. 18 (4) (2021) 046020.

[46] X. Shen, X. Liu, X. Hu, D. Zhang, S. Song, Contrastive learning of subject-invariant EEG representations for cross-subject emotion recognition, IEEE Trans. Affect. Comput. (2022).

[47] R.-N. Duan, J.-Y. Zhu, B.-L. Lu, Differential entropy feature for EEG-based emotion classification, in: 2013 6th International IEEE/EMBS Conference on Neural Engineering, NER, IEEE, 2013, pp. 81–84.

[48] P.L. Nunez, R. Srinivasan, Electric Fields of the Brain: The Neurophysics of EEG, Oxford University Press, USA, 2006.

[49] B. Crouch, L. Sommerlade, P. Veselcic, G. Riedel, B. Schelter, B. Platt, Detection of time-, frequency-and direction-resolved communication within brain networks, Sci. Rep. 8 (1) (2018) 1825.

[50] T. Koenig, F. Marti-Lopez, P. Valdes-Sosa, Topographic time-frequency decomposition of the EEG, Neuroimage 14 (2) (2001) 383–390.

[51] K. He, X. Zhang, S. Ren, J. Sun, Deep residual learning for image recognition, in: Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition, 2016, pp. 770–778.

[52] X. Hu, F. Wang, D. Zhang, Similar brains blend emotion in similar ways: Neural representations of individual difference in emotion profiles, Neuroimage 247 (2022) 118819.

[53] J. Chen, X. Wang, C. Huang, X. Hu, X. Shen, D. Zhang, A large finer-grained affective computing EEG dataset, Sci. Data 10 (1) (2023) 740.

[54] J. Bogaarts, D.M. Hilkman, E.D. Gommer, V. van Kranen-Mastenbroek, J.P. Reulen, Improved epileptic seizure detection combining dynamic feature normalization with EEG novelty detection, Med. Biol. Eng. Comput. 54 (2016) 1883–1892.

[55] D. Bollegala, Dynamic feature scaling for online learning of binary classifiers, Knowl.-Based Syst. 129 (2017) 97–105.

[56] J.J. Bird, J. Kobylarz, D.R. Faria, A. Ekárt, E.P. Ribeiro, Cross-domain MLP and CNN transfer learning for biological signal processing: EEG and EMG, IEEE Access 8 (2020) 54789–54801.

[57] B. Fernando, A. Habrard, M. Sebban, T. Tuytelaars, Unsupervised visual domain adaptation using subspace alignment, in: Proceedings of the IEEE International Conference on Computer Vision, 2013, pp. 2960–2967.

[58] J.P. Dmochowski, P. Sajda, J. Dias, L.C. Parra, Correlated components of ongoing EEG point to emotionally laden attention–a possible marker of engagement? Front. Hum. Neurosci. 6 (2012) 112.

[59] L.C. Parra, S. Haufe, J.P. Dmochowski, Correlated components analysis-extracting reliable dimensions in multivariate data, 2018, arXiv preprint arXiv:1801.08881.

[60] J. Wang, Cross-subject domain adaptation study for EEG-based emotion recognition, Neural Netw. Theory Appl. 14 (8) (2022) 1–7.

[61] Y. Ganin, E. Ustinova, H. Ajakan, P. Germain, H. Larochelle, F. Laviolette, M. March, V. Lempitsky, Domain-adversarial training of neural networks, J. Mach. Learn. Res. 17 (59) (2016) 1–35.

[62] N. Shen, et al., Contrastive learning of subject-invariant EEG representations for cross-subject emotion recognition, IEEE Trans. Affect. Comput. 14 (3) (2022) 2496–2511.

[63] T. Zhang, et al., GCB-Net: Graph convolutional broad network and its application in emotion recognition, IEEE Trans. Affect. Comput. 13 (1) (2019) 379–388.

[64] T. Song, et al., EEG emotion recognition using dynamical graph convolutional neural networks, IEEE Trans. Affect. Comput. 11 (3) (2018) 532–541.

[65] Y. Ding, et al., Tsception: Capturing temporal dynamics and spatial asymmetry from EEG for emotion recognition, IEEE Trans. Affect. Comput. 14 (3) (2022) 2238–2250.

[66] S. Zhang, C. Tang, C. Guan, Visual-to-EEG cross-modal knowledge distillation for continuous emotion recognition, Pattern Recognit. 130 (2022) 108833.

[67] Y. Xu, et al., AMDET: Attention based multiple dimensions EEG transformer for emotion recognition, IEEE Trans. Affect. Comput. (2023).

[68] E. Zanaty, Support vector machines (SVMs) versus multilayer perception (MLP) in data classification, Egypt. Inform. J. 13 (3) (2012) 177–183.

[69] Y. Li, J. Jin, H. Tao, Y. Xiao, J. Liang, C.P. Chen, Complemented subspace-based weighted collaborative representation model for imbalanced learning, Appl. Soft Comput. 153 (2024) 111319, http://dx.doi.org/10.1016/j.asoc.2024.111319.

[70] Y. Li, J. Jin, J. Ma, F. Zhu, B. Jin, J. Liang, C.P. Chen, Imbalanced least squares regression with adaptive weight learning, Inform. Sci. 648 (2023) 119541, http://dx.doi.org/10.1016/j.ins.2023.119541.

[71] M. Koklu, K. Sabancı, The classification of eye state by using kNN and MLP classification models according to the EEG signals, Int. J. Intell. Syst. Appl. Eng. 3 (4) (2015) 127–130.

[72] A. Mathur, G.M. Foody, Multiclass and binary SVM classification: Implications for training and classification users, IEEE Geosci. Remote Sens. Lett. 5 (2) (2008) 241–245.

[73] S. Dreiseitl, L. Ohno-Machado, Logistic regression and artificial neural network classification models: a methodology review, J. Biomed. Inform. 35 (5–6) (2002) 352–359.

[74] Z. Li, H. Zhang, Fusing deep metric learning with KNN for 12-lead multi-labelled ECG classification, Biomed. Signal Process. Control 85 (2023) 104849.